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International Academic Research Competition

Accessible Surface Treatments for Drag Reduction in Underwater Robots: A Graded Systematic Review of Reported Performance, Durability and Fabrication Accessibility

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Research Topic: Determining which surface treatments for reducing hydrodynamic drag on underwater robots hold up under realistic operating conditions and can be fabricated with equipment a small team can reach, by systematically reviewing and grading the published evidence.

Abstract

Underwater robots spend most of their stored energy pushing a hull through water, and on a slender hull most of that resistance is skin friction [1]. A surface treatment is the only way to reduce it without rebuilding the vehicle, the one option open to a group that owns a robot rather than designs one. The literature reports reductions from a few per cent to above thirty, measured in configurations that cannot be lined up. This study grades every claim on two axes the literature does not record: how directly the measurement bears on a working vehicle, and whether a small team could fabricate the surface at all. A PRISMA-guided search will assemble roughly ninety studies, and three pre-registered hypotheses predict that reported reduction falls as tests approach a real vehicle, that gas-dependent surfaces retain less benefit than geometric ones, and that most studies omit what is needed to reuse their result. Outputs are a graded shortlist, a retained-performance figure of merit, a reporting minimum and a bench-validation protocol.

Keywords: *hydrodynamic drag reduction; skin friction; underwater robots; riblets; superhydrophobic surfaces; lubricant-infused surfaces; fabrication accessibility; systematic review; evidence grading*

Introduction

20 Autonomous Underwater Vehicles and Remotely Operated Vehicles survey and inspect the sea, and their working range is set by the energy carried on board against the energy needed to overcome hydrodynamic resistance [1]. On a slender hull most of that resistance is skin friction. Hull shape and propulsion are fixed once a vehicle is built; the surface is not, which is why coatings and textures are the only drag reduction available to a group retrofitting a robot it already owns.

18 The laboratory literature is encouraging and not comparable. A skin-friction reduction on a flat plate, a total-drag reduction on a vehicle and a torque reduction on a rotating cylinder are different quantities. A percentage cannot be interpreted unless the quantity measured, the reference surface, the Reynolds number (a dimensionless measure combining flow speed, size and fluid properties) and the treated fraction of wetted area are stated.

A second question is barely asked. The best open-water result to date was measured on four-by-seven-centimeter silicon samples of re-entrant micro-trenches [2], which a group retrofitting a survey robot cannot make, while others are sprayed from a commercial formulation [3]. I could not find a synthesis that separates the two. This study grades the evidence on both axes at once (Figure 1).

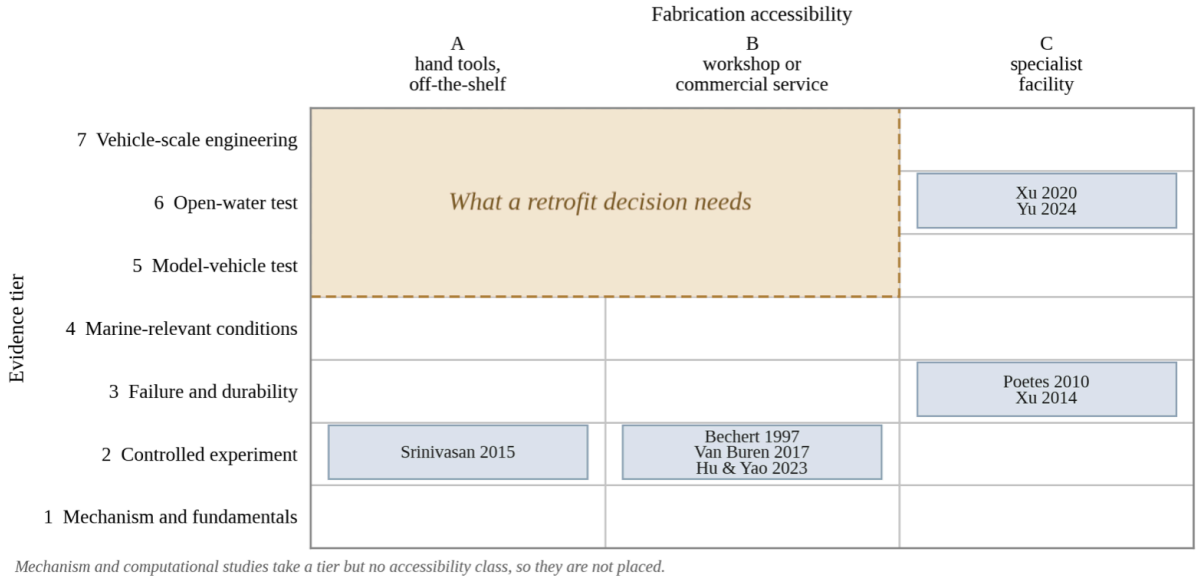


Figure 1. Where the cited evidence sits on the two grading axes. Placement is illustrative; coding the full sample is this study's output. In this placement the shaded region, the evidence a retrofit actually needs, holds nothing.

Research Question and Hypotheses

The research question is:

Among surface treatments a small team can fabricate and apply, which reduce hydrodynamic drag on an underwater robot by an amount that survives realistic operation, and how strongly does the published evidence support each one?

H1. Reported drag reduction falls as the test configuration approaches a working vehicle. Each claim is placed on a seven-tier evidence ladder running from mechanism studies to instrumented vehicles at sea, and magnitude is expected to decline as tier rises, because the largest values come from rotating rigs and small plates low on the ladder. Tested one-sided by Spearman rank correlation at the 5% level, restricted to claims reporting percentage reduction against a smooth reference, since pooling quantities would repeat the error named above. H1 is rejected if the correlation is not significantly negative.

H2. Treatments depending on a trapped gas or oil layer retain less of their initial benefit than treatments working by geometry alone. Among studies reporting performance after a stated duration or exposure, the retained fraction will be lower for superhydrophobic and lubricant-infused surfaces than for riblets and other geometric texturing. Tested one-sided by Mann-Whitney at the 5% level. Retention is reported rarely, so the subgroup may be small; the comparison is published with its sample sizes and is abandoned, not weakened, if either group holds fewer than eight studies.

H3. Fewer than one study in five reports all four core fields: the quantity measured, the reference surface, the Reynolds number with its definition, and the treated fraction of wetted area. The threshold follows from the rarest field, since a conjunction can be no commoner than its scarcest member, and the treated fraction is expected to be seldom reported. Tested one-sided at the 5% level: H3 is supported only if the 95% upper bound on that proportion falls below 0.20, and is otherwise not supported.

Research Aims and Objectives

The aim is a graded, usable account of which accessible treatments the evidence supports. The objectives are to assemble a pre-registered, PRISMA-screened sample; place every claim on the evidence ladder and the fabrication-accessibility class using frozen rules; extract reduction, retention and the conditions needed to interpret either; test the three hypotheses; and publish a shortlist, a reporting minimum and a bench-validation protocol.

Literature Review

Five families, and what each is reported to achieve

Riblets are stream-wise grooves working by geometry alone: Bechert et al. obtained a 9.9% skin-friction reduction with thin-blade riblets and 8.2% with a durable trapezoidal profile [4]. Superhydrophobic surfaces hold a plastron, a thin film of air the texture traps underwater; a sprayable coating gave up to 22% wall-shear reduction [3]. Lubricant-infused surfaces replace the air with oil, exceeding 35% in Taylor–Couette flow [5]. A sawtooth riblet with a superhydrophobic coat gave 20.2% where the coating alone was nearly ineffective [6]. Open-water tests are rarest: roughly 30% beneath a motorboat [2], and one at sea needing a pinned plastron throughout [7].

What decays, and how quickly

None of those numbers is a property of the coating alone. The plastron is metastable: it persists for a well-defined onset time then decays rapidly, with that time set by immersion depth [8], though air pockets survived beyond fifty days in engineered micro-trenches [9]. Benefit fell, or became a drag increase, as feature size grew, so contact angle does not predict performance [10]. Lubricant-infused surfaces relocate the problem: a two-phase Couette simulation finds that losing only 20% of the lubricant makes it superfluous [11]. Tang et al. identify material durability and plastron stability as the limits on adoption [12]. A percentage measured at the start of a run says nothing about the drag on hour six of a survey.

Why the reported percentages cannot be lined up

The decisive case is Kim et al., who simulated stream-wise V-riblets on a full-scale SUBOFF bare hull: an approximately 50% reduction in wall shear stress over the central region gave roughly 3% total-drag reduction when the riblet was correctly sized, and roughly 3% total-drag increase when oversized on the same hull [13]. One study, one surface, two numbers differing by more than an order of magnitude and opposite in sign. Zheng et al. collect values from single

digits to above 26%, each from a different primary study under different conditions [14], and Park, Choi and Kim call progress in turbulent flow rather tortuous, regular only around 2010 and successful in open water only in 2020 [15]. These reviews record obstacles and spread, but none grades a claim by how directly it bears on a vehicle, or records whether the surface could be made outside a cleanroom. The diagnosis assembled here is this study's own.

Grading evidence is a solved problem elsewhere

Other fields answered the same difficulty procedurally: PRISMA standardizes how a search and screen are reported [16], GRADE folds risk of bias, inconsistency, imprecision and indirectness into a single rating [17], and coder agreement is reported as weighted kappa [18], corrected for chance [19] and read against published bands [20]. The ladder proposed here isolates one axis of that machinery, indirectness of configuration. Applying these instruments to underwater drag-reduction evidence has not been done to the extent of my knowledge, so a novelty search forms part of the protocol.

Methodology

Approach and pre-registration: A systematic review with quantitative synthesis, run on published literature, needing no laboratory and no paid tool. Before screening begins, the protocol, instruments, hypotheses, thresholds and analysis plan are deposited on a public registry with a timestamp. Nothing changes afterwards without a dated amendment, which stops the thresholds moving once results are visible.

Search and sources: Searches run through OpenAlex [21], Crossref and Semantic Scholar, all free. Query families pair a mechanism term with a condition term, and backward and forward citation chasing runs on every included study. Where Scopus or Web of Science access exists it is added as a coverage check.

Eligibility and screening: Eligible records report an original quantitative drag, friction, torque or distance result for a surface treatment tested in water or a marine setting. Reviews, editorials, wettability-only studies and unvalidated simulations are excluded. Screening follows PRISMA [16], with counts at each stage and a published flow diagram.

The two grading instruments: The evidence ladder has seven tiers, ranking a claim by how directly its measurement bears on a decision about a real vehicle, as Figure 1 sets out. The fabrication-accessibility class scores four axes separately: equipment, consumable cost per square meter, whether it can be applied to an existing hull, and maintenance burden. Scores combine into the three classes Figure 1 names, and apply only to a physically fabricated surface. Both instruments are piloted on fifteen held-out studies, and the codebook is frozen and registered before coding.

Data extraction: One primary claim per study is coded by a written rule: the result the abstract leads with. Fields recorded are texture dimensions, fabrication route and cost, treated fraction of

wetted area, flow conditions with the Reynolds-number definition, which quantity was measured against which reference, and performance retained after a stated duration.

Coding reliability: Tier and accessibility class are judgements, so they are checked rather than asserted. A second reader, trained on the codebook but not involved in writing it, independently codes a 30% random subsample blind to the first coder's entries. Linearly weighted kappa is reported for tier and unweighted kappa for accessibility class [18], [19], read against the Landis and Koch bands [20]. Agreement below 0.60 halts the analysis: the codebook is revised, the revision is logged as an amendment, and the subsample is recoded. Disagreements are logged with their cause, whether information absent from the paper, ambiguity in the rule, or coder error.

Analysis: H1 is tested by one-sided Spearman rank correlation between tier and reported reduction. H2 compares retained fraction between gas-dependent and geometric treatments by one-sided Mann-Whitney. H3 reports the proportion carrying all four core fields with a Wilson interval, which behaves better than a normal approximation near zero.

Sample size: The target of ninety studies is set by the coding effort available. Ninety pairs would detect a Spearman correlation of 0.30, the smallest association that would matter here, at the 5% one-sided level with power near 0.88. H1 uses only the subset reporting percentage reduction against a smooth reference, so its real power is lower: sixty such claims give about 0.73 and fifty about 0.66. The subset cannot be sized before screening, so achieved power is reported with the result.

Practicalities and timeline: The study needs a computer, free database interfaces, a spreadsheet and one trained second reader. Coding is the largest cost, roughly fifty person-hours across both readers. Months 1 to 2 draft, pilot and pre-register the instruments; month 3 runs the searches and screening; months 4 to 5 cover coding and the reliability subsample; month 6 is analysis and release of the codebook and dataset.

Risks and limitations: Cost and equipment judgements carry uncertainty, so the four accessibility axes are published separately. The review establishes what the evidence supports, not what a treatment will do on a particular hull.

Expected Outcomes

Four outputs result whichever way the hypotheses fall: an open dataset of coded claims with its PRISMA flow; a graded shortlist of class A and B treatments, each carrying the tier of its evidence; a figure of merit the field cannot compute today, retained drag reduction at a stated cost; and a reporting minimum.

Conclusion and Post-Program Plan

The result is meant to be used: a shortlist a small group can act on, and a protocol for testing it. The coded dataset, codebook and disagreement log will be released under an open license, and the results submitted to a peer-reviewed journal. The next study is the bench-validation protocol this review produces.

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